

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Priyanshi Halpani
Program/Batch	B082-Networkwalks
Date	20 th August 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM3 (Footprinting with Maltego) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on systems and devices for which I had proper authorization or that I personally own. These activities were carried out strictly for educational and research purposes as part of my cybersecurity training.

The information and techniques presented in this report should not be used for unauthorized access, testing, or any illegal activity. I take full responsibility for the activities performed and the use of the knowledge gained from this practical work.

Any reconnaissance, scanning, or security testing should always be conducted within an authorized scope and with appropriate permission. Unauthorized access or testing of computer systems and networks may have legal and professional consequences.

2. Introduction

This report covers footprinting and reconnaissance of the networkwalks.com domain using multiple Kali Linux tools and **Maltego**, along with network scanning of my own local network using **Nmap/Zenmap**. The activities cover the footprinting and scanning phases, demonstrating how publicly available information can be collected and correlated before identifying live hosts and services on a network. This report documents the Week 2 activities completed as part of my ongoing internship program at Networkwalks.

All footprinting activities were performed in Kali Linux, while network scanning was conducted using Nmap/Zenmap on a Windows PC. The tools were used to collect domain, web technology, DNS, WAF, email and network information. Maltego was additionally used to visualize relationships between the domain and publicly available information. Each activity includes the command or procedure used, the observed result, supporting screenshots, and a brief explanation of the security relevance of the finding.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Maltego	Visualize relationships between domains, email addresses and other publicly available entities during reconnaissance.
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance on **networkwalks.com** using six Kali Linux tools: **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f**, and **DNSRecon**, along with **Maltego** for visualizing relationships between the domain and associated email information.

WHOIS provided domain registration details, while WhatWeb identified technologies including **WordPress 7.0.4**, **WordPress Download Manager 3.3.58**, **Apache**, and **jQuery 3.7.1**. Nslookup resolved the domain to **192.232.216.135**, and Curl revealed HTTP headers and the **/wp-json/** WordPress REST API endpoint. Wafw00f identified **ModSecurity (SpiderLabs)** as the WAF, while DNSRecon provided information about **DNS**, **mail**, **SPF/TXT**, and **service records**.

Finally, Maltego was used to visually correlate the domain with publicly available email information. All activities were conducted for authorized information gathering and analysis, with no exploitation or unauthorized access performed.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The example results provided in the practical identified three live hosts:

- 192.168.201.1
- 192.168.201.100
- 192.168.201.103

The example results also included four MAC addresses.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

Risk Analysis / Impact				
#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
 1	Web technology and version information exposed	WhatWeb identified Apache, WordPress 7.0.4, WordPress Download Manager 3.3.58, jQuery 3.7.1 and other web technologies.	Exposed technology and version information may help an attacker identify the technologies used by the website and determine whether further security review is required.	• Medium
 2	Server IP address publicly identifiable	WhatWeb and nslookup identified 192.232.216.135 as the IP address associated with network-walks.com.	Knowledge of the server IP can provide information about the hosting infrastructure and assist further reconnaissance.	• Low
 3	HTTP response information exposed	Curl returned HTTP response headers including Server: Apache, WordPress-related headers and links to the WordPress REST API endpoint /wp-json/.	Technical information exposed through HTTP headers may assist technology fingerprinting and further enumeration.	• Low
 4	WordPress REST API endpoint exposed	The HTTP Link header referenced https://network-walks.com/wp-json/ and a WordPress REST API resource.	Publicly accessible API information may provide additional information about the WordPress installation and should be reviewed for unnecessary information exposure.	• Low
 5	Web Application Firewall identified	Wafw00f identified ModSecurity (SpiderLabs) protecting the website.	Identification of the WAF provides information about the security architecture that may be useful during further reconnaissance.	• Low

	DNS infrastructure information exposed	DNSRecon identified SOA, NS, MX, A, TXT and SRV records, including HostGator name servers, mail infrastructure and SPF information.	DNS information can help build a broader profile of the organization's hosting, email and service infrastructure.	• Medium
	DNS server software/version information exposed	DNSRecon identified the Bind version as 9.16.23-RH for the discovered name servers.	Exposed software/version information may assist attackers in identifying the technologies used in the DNS infrastructure and determining whether further review is required.	• Medium
	Multiple network services exposed on an internal host	Nmap identified 135/tcp, 139/tcp, 445/tcp and 9999/tcp as open on 192.168.201.103.	Multiple exposed services increase the network attack surface and should be reviewed to determine whether they are required and properly secured.	• Medium
	HTTP service exposed on local network host	Nmap identified 80/tcp (HTTP) as open on 192.168.201.1.	An accessible HTTP service may expose a device management interface or other web service and should be reviewed for secure configuration.	• Medium
	Multiple active hosts identified on local network	Nmap scanned 256 IP addresses and identified 3 hosts as up: 192.168.201.1, 192.168.201.100 and 192.168.201.103.	Identifying active devices provides useful information about the network structure. Unknown or unauthorized devices could represent a security concern.	• Low

Made with  Napkin

The risks above are observations from the footprinting, reconnaissance and network-scanning exercises and should not be interpreted as confirmed vulnerabilities.

The activities primarily involved passive/public information gathering, service identification and network discovery. Although several technologies, services and versions were identified, the results alone do not confirm that any of them are vulnerable.

For example, the identification of WordPress 7.0.4, Apache, Bind 9.16.23-RH or the open ports discovered by Nmap indicates technologies or services that are present. Further authorized vulnerability assessment would be required to determine whether these components contain exploitable security weaknesses.

The Nmap scan identified three active hosts within the scanned 256-address range. The host 192.168.201.103 exposed ports 135, 139, 445 and 9999, while 192.168.201.1 exposed port 80. These

findings represent an increased attack surface but do not by themselves demonstrate successful exploitation.

6. Recommendations

1. Based on the observations from the footprinting, reconnaissance and network-scanning activities, I recommend the following security improvements:
2. **Review publicly exposed technology information**
Organizations should regularly review publicly visible information about their web server, CMS, plugins and JavaScript libraries. Where possible, unnecessary version information should not be exposed.
3. **Keep web technologies updated**
WordPress, WordPress Download Manager, Apache, jQuery and other components should be regularly updated and reviewed against current security advisories.
4. **Review HTTP response headers**
HTTP response headers should be reviewed to determine whether unnecessary technical information is being disclosed. Security-related headers should also be configured appropriately.
5. **Review the WordPress REST API exposure**
The /wp-json/ endpoint should be reviewed to determine what information is publicly accessible and whether unnecessary information is exposed.
6. **Review DNS records regularly**
DNS records should be periodically reviewed to ensure that only required records and services are publicly exposed.
7. **Review DNS server configuration**
The identified DNS infrastructure and software versions should be reviewed and kept updated. Unnecessary information disclosure should be minimized.
8. **Properly configure and monitor the WAF**
ModSecurity should remain enabled and properly configured. WAF rules and logs should be monitored to identify suspicious or repeated requests.
9. **Review open ports on internal hosts**
The services exposed on 192.168.201.103, particularly ports 135, 139, 445 and 9999, should be reviewed to determine whether they are required.
10. **Restrict unnecessary network services**
Firewalls and access-control mechanisms should be used to restrict services such as SMB-related ports to trusted systems where they are required.
11. **Perform regular internal network discovery**
Authorized network scans should be performed periodically to identify active devices, newly exposed services and unexpected changes in the network.
12. **Investigate unknown devices**
Any unidentified or unauthorized host discovered during network scanning should be investigated and verified.

13. Maintain network documentation

IP addresses, devices, services and network topology should be documented and regularly updated.

14. Perform further security testing with authorization

Any vulnerability assessment or exploitation attempt should only be performed against systems for which appropriate authorization has been obtained.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting and reconnaissance activity, I used multiple Kali Linux tools to collect information about the networkwalks.com domain. WHOIS provided domain registration information, including the registrar, registration dates and domain status. WhatWeb identified technologies such as Apache, WordPress 7.0.4, WordPress Download Manager 3.3.58 and jQuery 3.7.1.

Using nslookup, I resolved networkwalks.com to the IP address **192.232.216.135**. Curl was then used to inspect the HTTP response headers and identify additional technical information, including the Apache server and the WordPress REST API endpoint.

Wafw00f identified **ModSecurity (SpiderLabs)** as the Web Application Firewall protecting the website. DNSRecon provided additional information about the DNS infrastructure, including SOA, NS, MX, A, TXT and SRV records, as well as DNS server software information. I also used **Maltego** to visualize and correlate the relationship between the networkwalks.com domain and publicly available email information, helping me understand how reconnaissance data can be connected and represented graphically.

For the network-scanning activity, I used Nmap to scan a local network containing **256 IP addresses**. The scan identified **three active hosts**: 192.168.201.1, 192.168.201.100 and 192.168.201.103.

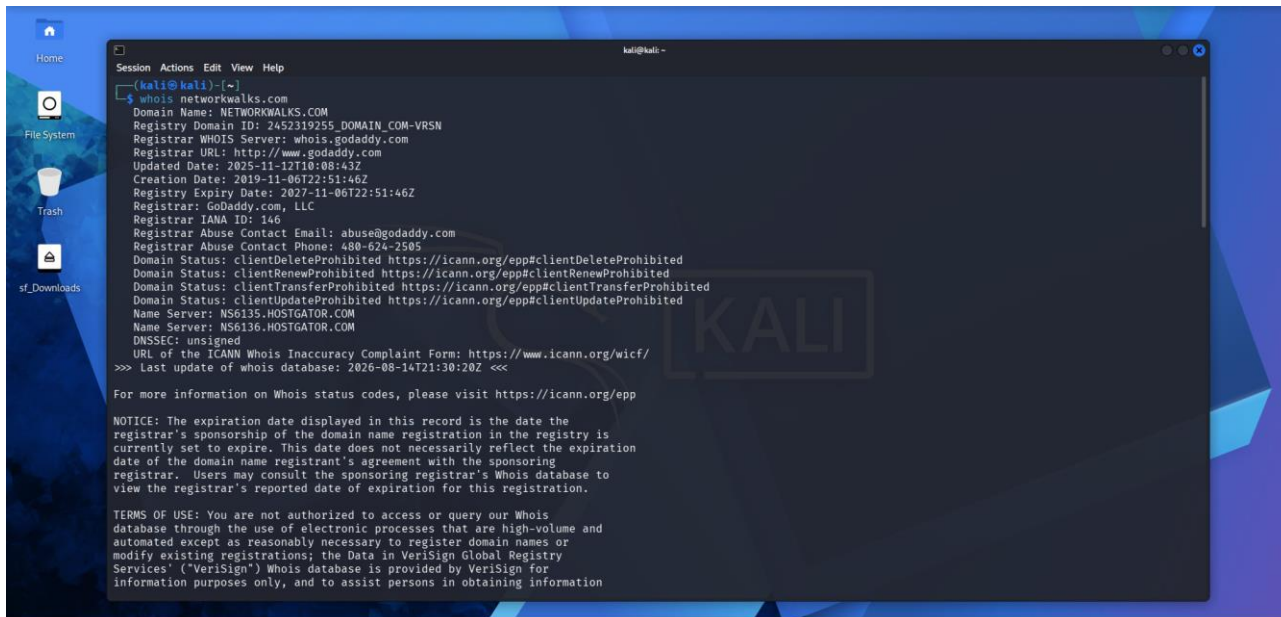
The scan showed that 192.168.201.1 had port **80/tcp** open, while 192.168.201.103 had ports **135/tcp**, **139/tcp**, **445/tcp** and **9999/tcp** open. The host 192.168.201.100 was identified as active, but all 100 scanned TCP ports were reported as closed.

Through these activities, I learned how reconnaissance can be used to collect and correlate information about a web application's technologies, DNS infrastructure, server configuration and network environment. I also learned how Maltego can help visualize relationships between different pieces of publicly available information.

The exercises demonstrated the importance of distinguishing between a **security observation** and a **confirmed vulnerability**. Identifying a software version, DNS record or open port does not automatically mean that the system is vulnerable. Further authorized testing would be required to validate any potential security weakness.

Finally, I learned that reconnaissance and network scanning must always be performed within an authorized scope. The activities documented in this report were performed as part of the assigned educational cybersecurity practical.

8. Evidences Collected

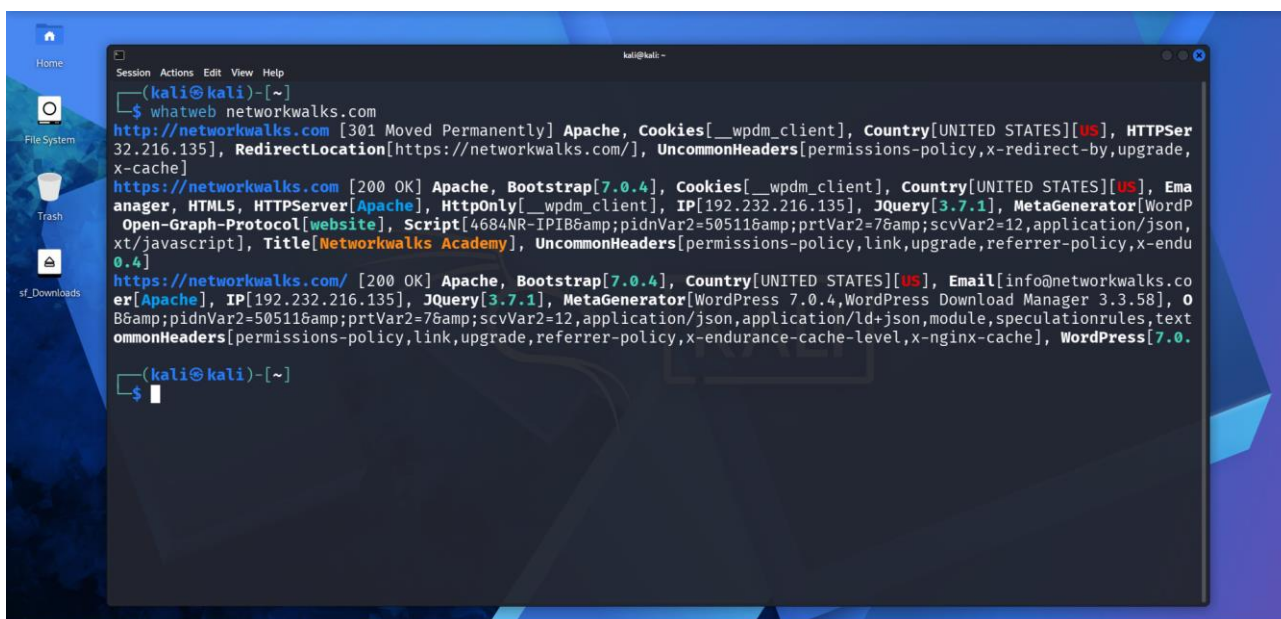


```
(kali@kali)-[~]
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-08-14T21:30:20Z <<<

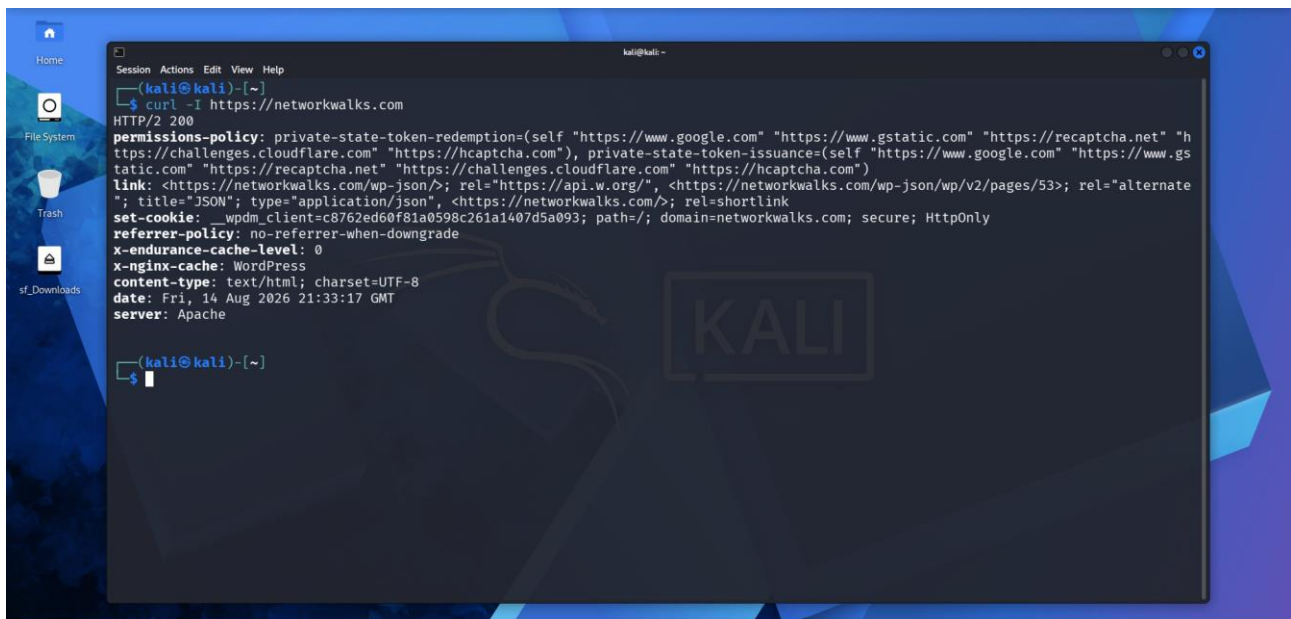
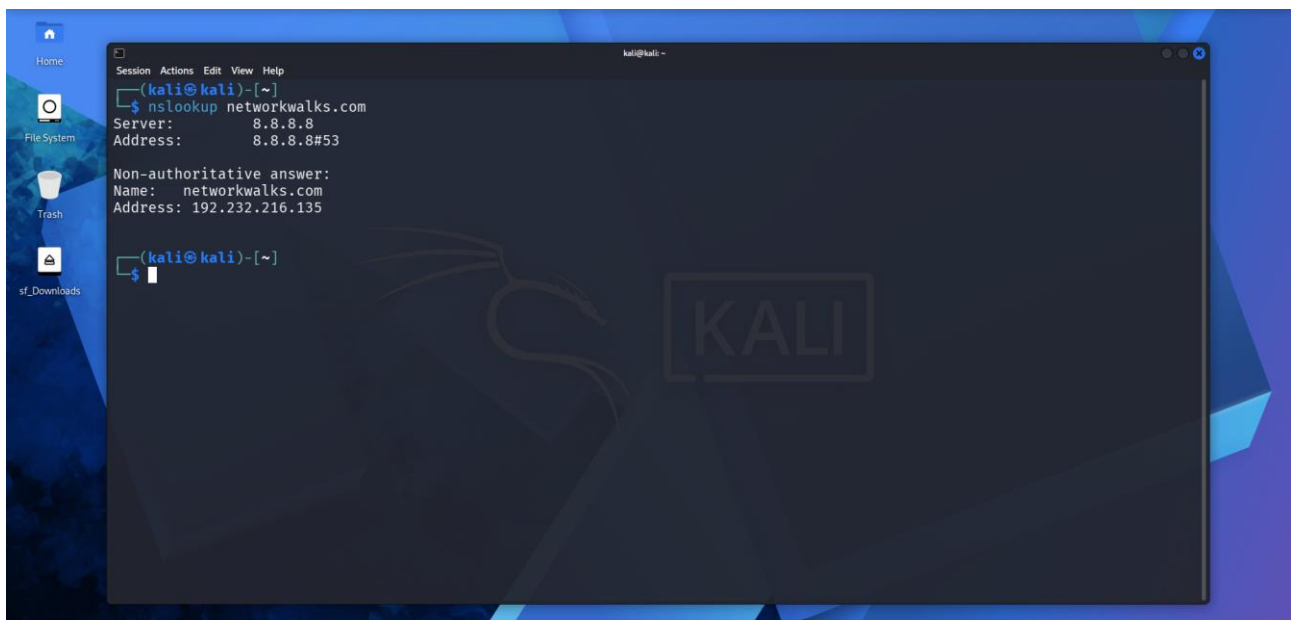
For more information on Whois status codes, please visit https://icann.org/epp

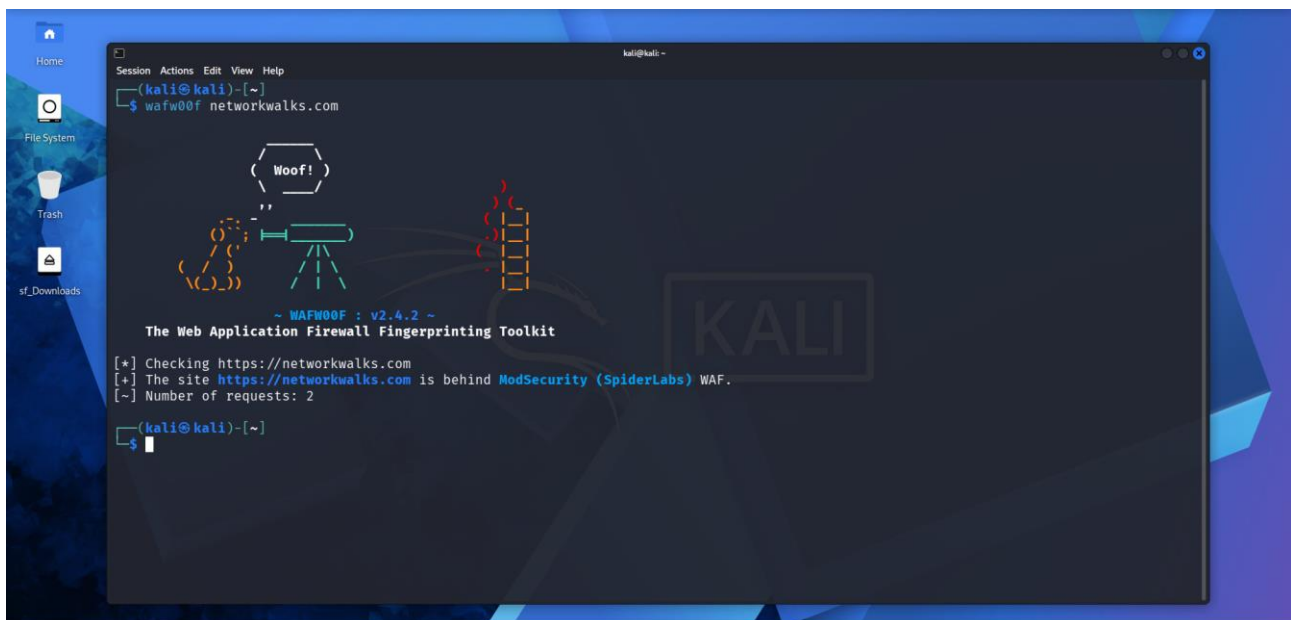
NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' (VeriSign) Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
```



```
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer
32.216.135, RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,
x-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.0.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Ema
anager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordP
Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,
xt/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endu
0.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.0.4], Country[UNITED STATES][US], Email[info@networkwalks.co
er[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Manager 3.3.58], O
B&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text
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```

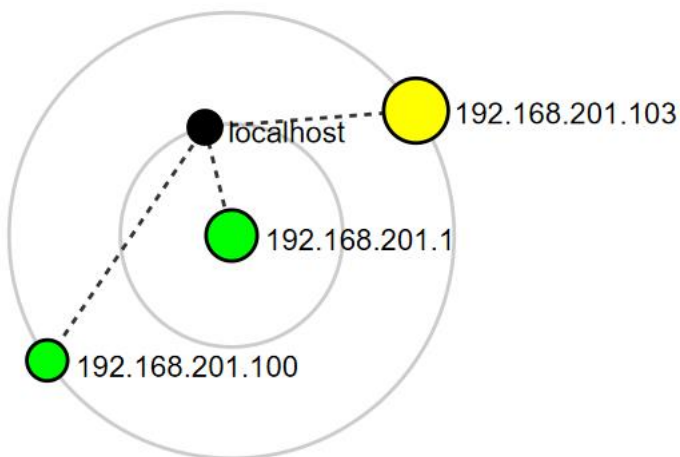
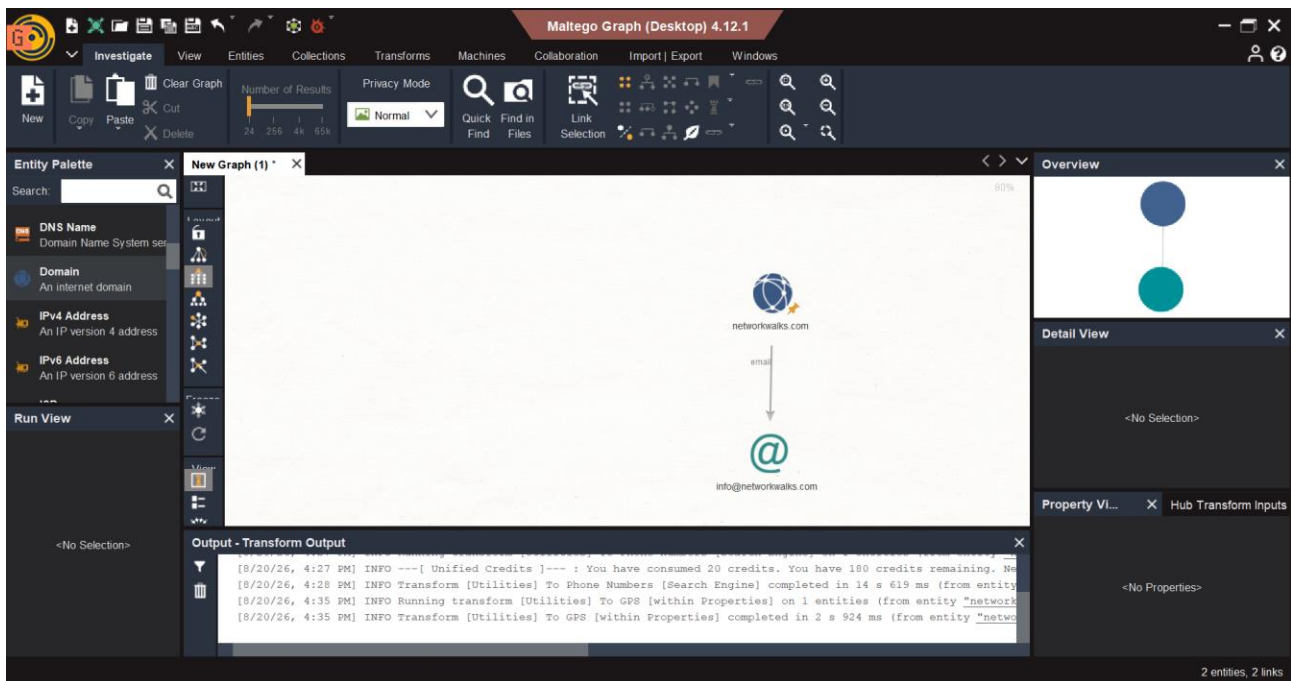





```

~(kali@kali)-[~]
$ dnsrecon -d networkwalks.com
2026-08-18T07:43:52.668081-0400 INFO Starting enumeration for domain: networkwalks.com
2026-08-18T07:43:52.669104-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-08-18T07:43:53.279660-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-08-18T07:43:53.373244-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-08-18T07:43:53.472608-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-08-18T07:43:54.136820-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-08-18T07:43:54.137196-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-08-18T07:43:54.713257-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-08-18T07:43:55.633844-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-08-18T07:43:55.651001-0400 INFO A networkwalks.com 192.232.216.135
2026-08-18T07:43:56.557524-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-08-18T07:43:56.557860-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-08-18T07:43:57.377640-0400 INFO Enumerating SRV Records
2026-08-18T07:44:03.682596-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-08-18T07:44:03.683260-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-08-18T07:44:03.683455-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-08-18T07:44:03.684756-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-08-18T07:44:03.685149-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-08-18T07:44:03.685388-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-08-18T07:44:03.685672-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-08-18T07:44:03.685928-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-08-18T07:44:05.003579-0400 INFO 8 Records Found
2026-08-18T07:44:05.004138-0400 INFO Completed enumeration for domain: networkwalks.com

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-End-



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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub

